i-CAPE

ISO

Innovative Science Olympiad

2026-2027

Grade 6

Total Questions: 35 | LEVEL – II | Set - B | Time: 1.5 hr.

Total Marks: 70

SECTION - 1	• This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 2	• This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s). • For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. • There are **NO NEGATIVE** marks for wrong answers.
SECTION - 3	• This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. • Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 4	• This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS.** • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A student took three test tubes and added the following:
Tube A: Mashed potato + Iodine solution.
Tube B: Crushed dal (pulses) + Copper sulphate solution + Caustic soda.
Tube C: Boiled rice (chewed for 2 minutes) + Iodine solution. Which of the following observations is scientifically correct?

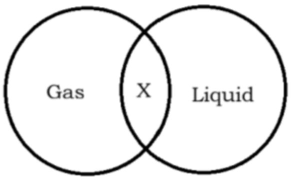
A) Tube A and Tube C will both turn blue-black immediately.

B) Only Tube B will show a violet color, indicating the presence of proteins.

C) Tube C will not turn blue-black because the saliva broke down the starch into sugars.

D) Tube A will turn red, indicating the presence of fats.

2. **Observe the Venn diagram given below.**



Which of the following properties belong to the X region?

(i) Have fixed volume

(ii) Can flow (iii) No fixed shape

(iv) Particles are very far apart

A) (i) only
B) (ii) only
C) (ii) and (iii) only
D) (i) and (iii) only

3. **A student uses a laboratory thermometer to measure temperature. He removes the thermometer before noting the reading. What problem may occur?**

A) Reading cannot be retained after removal
B) Temperature increases suddenly
C) Thermometer breaks
D) Reading becomes higher

4. **Growth is a characteristic of all living things. However, a crystal of sugar kept in a saturated solution also "grows" in size over time. How is the growth of a living organism different from the "growth" of a sugar crystal?**

A) There is no difference; both are considered living.
B) Living growth is from within (internal), while crystal growth is just the addition of material on the outside (external).
C) Crystals grow only at night, while living things grow during the day.
D) Living things grow only until a certain age, but crystals grow forever.

5. **Assertion (A): Trees in mountain habitats are usually cone-shaped and have sloping branches.**

Reason (R): This shape helps the rainwater and snow to slide off easily, preventing damage to the tree.

A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is not the correct explanation of A.
C) A is true but R is false.
D) A is false but R is true.

6. **A soft iron rod is placed near a magnet and then hammered while still in the magnetic field. What will happen?**

A) It becomes a stronger temporary magnet
B) It becomes completely demagnetised
C) It turns into a permanent magnet instantly
D) It shows no magnetic effect

7. **Consider the following statements about compounds and mixtures.**

P. All compounds are homogeneous in nature.

Q. The constituents of a mixture cannot be separated by physical methods.

R. The composition of a compound is in a fixed ratio by weight.

S. The components of a mixture retain their individual properties.

Select the correct combination of true (T) and false (F) statements.

	P	Q	R	S
A)	T	F	T	T
B)	T	T	F	T
C)	F	F	T	T
D)	T	F	F	T

8. Which does the below diagram represent?

A) Air consists of water vapour
B) Air consists of oxygen
C) Air consists of nitrogen
D) Air consists of carbon dioxide

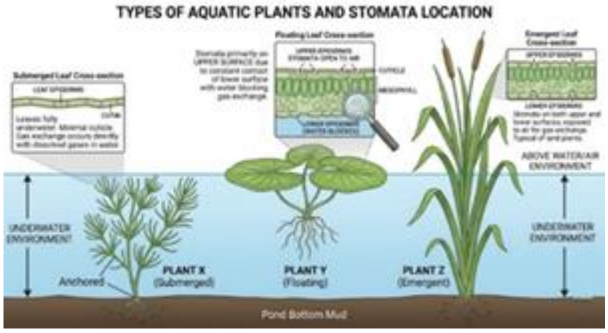
9. During a school health camp, four students shared their daily lunch menus.
Student 1: Rice, Dal, Ghee, Salad.
Student 2: Burger, French fries, soft drink.
Student 3: Steamed vegetables, Sprouted seeds, Curd, Chapati.
Student 4: Only fruits and water. Who among them is consuming a "Balanced Diet" that covers all essential nutrients including roughage?

A) Student 1 B) Student 2
C) Student 3 D) Student 4

10. A tree forms an image of height 30 cm in a pinhole camera. The screen is 10 cm from the pinhole, and the tree is 5 m away. Find the actual height of the tree.

A) 10 m B) 15 m
C) 12 m D) 8 m

11. [Imagine a diagram of three plants: X (Submerged), Y (Floating with broad leaves), and Z (Emergent)]. In which of these plants would you expect to find stomata primarily on the upper surface of the leaves, and why?



A) Plant X, because it is completely underwater.
B) Plant Y, because the lower surface is in constant contact with water, which would block gas exchange.
C) Plant Z, because it needs to breathe more air than the others.
D) None of them have stomata because they live in water.

12. Study the given table below.

Planet	**Incoming Solar Radiation**	**Outgoing Radiation**	**Atmosphere**
Planet A	Allows	Traps	Present
Planet B	Reflects	Releases	Present

Which planet will be hotter and what is the reason behind it?

A) Planet A, Greenhouse effect
B) Planet B, Gravitational effect
C) Planet B, Greenhouse effect
D) Planet A, Reflection of sunlight

13. An earthworm is placed on a moist paper towel. A small drop of vinegar (acid) is placed near its front end. The earthworm immediately recoils and moves in the opposite direction. This experiment demonstrates that:

A) Earthworms hate the smell of vinegar.
B) Living organisms' sense and respond to chemical changes in their environment.
C) Earthworms move faster when they are hungry.
D) Vinegar provides energy to the earthworm's skin.

14. When performing the "Paper Patch Test" for fats, a student notices a spot on the paper after rubbing a piece of apple on it. However, after 10 minutes of waiting, the spot disappears. What does this indicate?

A) The spot was caused by water, not fat. A true fat spot is permanent and translucent.
B) The apple contains a "disappearing fat."
C) The paper absorbed the fat into its fibers.
D) The apple has more vitamins than minerals.

15. A person walks 100 m east in 20 s, then 60 m north in 15 s, then 100 m west in 20 s. Finally, he walks directly back to the starting point in 25 s. Find the average speed and average velocity respectively.

A) 8 m/s, 0 m/s
B) 4 m/s, 0 m/s
C) 8 m/s, 2 m/s
D) 10 m/s, 2 m/s

Section - 2

16. A scientist observes a plant with thin, ribbon-like leaves that are submerged in a freshwater stream. Why does the plant have this specific leaf structure?

A) To increase the surface area for absorbing sunlight under water.
B) To allow water to flow through them easily without damaging the plant.
C) To store excess air to help the plant float.
D) To reduce water loss through transpiration.

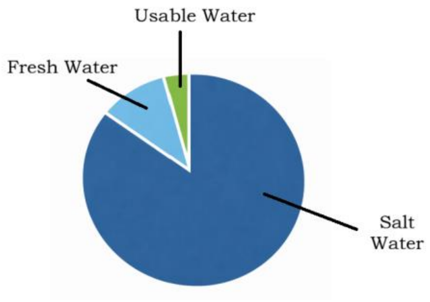
17. A point light source, an opaque object, and a screen are arranged in a straight line. The object is slowly moved towards the screen. Which of the following observations are correct?

A) Shadow size continuously decreases
B) Shadow size continuously increases
C) Rate of increase in shadow size is not uniform
D) Shadow disappears completely at some point

18. Most people think "movement" means running or walking, but in biology, it is much broader. Which of the following statements accurately describe movement as a characteristic of living creatures?

A) All living organisms must move from one geographical location to another to find food.
B) Plants exhibit movement by tracking the sun or opening and closing their petals.
C) Internal movement occurs in all living things, such as the flow of water and nutrients through specialized tissues.
D) Microscopic organisms like bacteria can use hair-like structures to move in fluids.

19. **Based on the composition of water on Earth shown in the pie graph, which of the following statements are correct?**



A) More than 95% of Earth's water is saline.
B) Fresh water constitutes less than 5% of total water.
C) All fresh water available on Earth is usable.
D) Usable water is only a very small fraction of total water.

20. **Different animals have evolved diverse organs to extract oxygen from their specific environments. Which of the following are scientifically accurate?**

A) Earthworms use their moist skin for gas exchange; if their skin dries out, they can no longer "breathe."
B) Frogs are unique because they can breathe through their lungs on land and through their skin while in water.
C) Insects breathe through a network of tiny tubes called tracheae, which open to the outside through holes called spiracles.
D) All animals that live in the ocean, including whales and dolphins, must have gills to breathe underwater

21. **Two planets: Planet X – rocky, close to Sun Planet Y – gaseous, far from Sun Which are correct?**

A) X is likely an inner planet
B) Y is likely an outer planet
C) X must always be hotter than Y
D) Y may have many moons

22. **Water and Roughage provide 0 calories (no energy). Which of the following are logical reasons why they are still categorized as "Components of Food"?**

A) Water is the primary medium (solvent) in which all life-sustaining chemical reactions in our cells take place.
B) Water helps in maintaining body temperature by absorbing heat and releasing it through sweat.
C) Roughage is digested by the body to provide a slow-release form of high-energy protein.
D) Without water, the body would be unable to transport nutrients to cells or flush out toxic wastes via urine.

23. **Which of the below diagram represent homogeneous mixture?**

A) 
Orange Juice with Pulp

B) 

Saline Water

C) 
Sand in Water

D) 
Sugar Water

Section - 3

COMPREHENSION-1

A student places a bar magnet near a compass. The compass needle shows a strong deflection. The magnet is slowly moved away, and the deflection decreases gradually. At a certain distance, the compass needle aligns exactly along the north–south direction.

Later, iron filings are sprinkled near the magnet. The filings stick more strongly at the ends and less at the center. The magnet is then heated and hammered, after which the filings stick weakly and are more spread out.

24. Why does deflection decrease gradually as magnet is moved away?

A) Magnet becomes smaller
B) Magnetic effect decreases with distance
C) Compass becomes weak
D) Earth's field disappears

25. Why does compass finally align north–south?

A) Magnet disappears
B) Earth's magnetic field dominates
C) Compass resets itself
D) Needle stops moving

26. Why do iron filings stick more at the ends?

A) Ends are longer
B) Poles have maximum strength
C) Center repels filings
D) Gravity pulls filings

COMPREHENSION-2

Roots absorb K from the soil, while leaves take in carbon dioxide. In the presence of sunlight and chlorophyll, K reacts with carbon dioxide to form food and oxygen. Pure K is a poor conductor of electricity, but when L are dissolved in it, it conducts electricity. Due to its ability to dissolve many substances, K is called M.

27. Identify M from the scrambled options.

A) VURSELIUN SOLTVEN
B) LAVIRSUN TOLSVAN
C) VERSALUNI TOLSVEN
D) VERSELUN SOLTVEN

28. A sample of K is cooled from room temperature to very low temperature. At which stage will K change into a solid under normal atmospheric pressure?

A) 100°C
B) 0°C
C) 50°C
D) It never solidifies

29. Two samples are taken: Sample 1: Pure K Sample 2: K with L dissolved in it When tested with a circuit, the bulb glows only in Sample 2. What is the most appropriate reason?

A) K itself is a good conductor
B) L increases the temperature of K
C) L provides charged particles that help in conduction
D) K evaporates in Sample 1

COMPREHENSION-3

Living organisms maintain internal stability despite fluctuating external conditions through a process known as homeostasis. This involves precise regulation of temperature, water balance, and chemical composition within the body. For example, humans sweat to lower body temperature, while certain aquatic organisms regulate ion concentration to survive in varying salinities. These mechanisms require coordinated functioning of multiple organ systems and are essential for sustaining life processes under environmental stress.

30. Homeostasis primarily ensures:

A) External environmental stability
B) Internal balance within the organism
C) Rapid growth under all conditions
D) Increased reproduction rate

31. Sweating in humans is mainly associated with:

A) Excretion of salts only
B) Regulation of body temperature
C) Digestion of food
D) Oxygen transport

32. Failure of homeostasis would most likely lead to:

A) Improved adaptation
B) Internal imbalance and dysfunction
C) Faster evolution
D) Increased immunity

Section - 4

33. Match Column A with Column B

Column A (Mixtures)	**Column B (Techniques Needed for Separation (Water may be added if required))**
(A) Ammonium chloride and table salt.	(P) Threshing, evaporation and sublimation.
(B) Harvested crop (grains + stalks) mixed with alcohol-water mixture	(Q) Sublimation, filtration and evaporation
(C) Harvested crop (grains + stalks), ammonium chloride, and table salt	(R) Distillation and threshing
(D) Ammonium chloride, sand, and table salt	(S) Sublimation

A) A–P, B–Q, C–R, D–S
B) A–Q, B–P, C–R, D–S
C) A–S, B–R, C–P, D–Q
D) A–S, B–Q, C–R, D–P

34. Match Column A with Column B

Column A	**Column B**
(A) Vitamin A	(P) Scurvy
(B) Vitamin C	(Q) Goitre
(C) Vitamin D	(R) Night Blindness
(D) Iron	(S) Rickets
(E) Iodine	(T) Anemia

A) A–R, B–P, C–S, D–T, E–Q
B) A–P, B–R, C–S, D–Q, E–T
C) A–R, B–Q, C–P, D–T, E–S
D) A–S, B–P, C–R, D–T, E–Q

35. Match Column A with Column B

Column A (Case)	**Column B (Correct Combined Concept)**
(A) Object returns to initial point after motion	(P) Displacement = 0, distance > 0
(B) Object moves in straight line in fixed direction	(Q) Velocity constant, displacement = distance
(C) Object changes direction continuously in motion	(R) Velocity changes even if speed constant
(D) Object covers equal distances in unequal times	(S) Average speed must be calculated

(A) A–P, B–Q, C–R, D–S
(B) A–Q, B–P, C–S, D–R
(C) A–P, B–R, C–Q, D–S
(D) A–S, B–Q, C–R, D–P